SLAC1 (Q9LD83) as a Potassium Channel Inhibitor (KAT1/AKT2) — Hypothesis Evaluation

Gene: SLAC1 (Arabidopsis thaliana, UniProt Q9LD83, AT1G12480) **Term under test:** GO:0019870 potassium channel inhibitor activity (MF) **Seed reference:** Zhang et al. 2016, Plant Cell 28(4):949-955 (P 27002025)

Hypothesis slug: slac1-kat1-inhibition

Executive Judgment

Verdict: Partially supported (weak–moderate), not independently replicated.

The seed hypothesis faithfully represents what Zhang et al. 2016 report, and that single paper provides genuinely direct evidence for the molecular activity: SLAC1 physically interacts with KAT1, KAT2 and AKT2 (Y2H, co-IP) and, in Xenopus oocyte voltage-clamp, **functionally inhibits the inward-K^+ (K^+in) channels KAT1 and AKT2 but not KAT2**, with guard-cell gain-of-function (overexpression) reducing K^+in currents and light-induced stomatal opening. This is sufficient to qualify SLAC1 as a candidate potassium-channel inhibitor (GO:0019870).

However, the stronger clauses of the hypothesis — "independently replicated or corroborated and not contradicted by later work" — are **not met**:

1. **No independent functional replication.** Searching the 45 PubMed "cited-by" records (2017–2026) found no independent lab reproducing SLAC1→KAT1/AKT2 patch-clamp inhibition. Citations are predominantly reviews or tangential.
2. **Database silence on the K^+-channel interaction.** IntAct curates 19 SLAC1 interactions, but **all are signaling kinases/phosphatases** (CPK6/21/23, PP2CA, SnRK2s/OST1) — **none with KAT1, AKT2 or KAT2**. The Zhang 2016 channel interaction has not been captured by interaction databases.

3. **The term is not currently annotated — this is an ADDITION.** UniProt/TAIR/ QuickGO carry no potassium-channel inhibitor (GO:0019870) or regulator (GO:0015459) MF term for SLAC1; current MF terms are anion-channel activity and kinase/phosphatase binding. So the curation question is whether to *add* a new MF annotation on single-study evidence, not whether to retain one.
4. **Competing/contradicting in-planta directionality.** Laanemets et al. 2013 (P 23126621) show that *slac1* loss-of-function **reduces** guard-cell K^+in and slows opening — the *opposite* of the "relief of inhibition" predicted by a direct-inhibition model — and attribute it to elevated cytosolic [Ca^{2+}] and increased Ca^{2+} sensitivity of K^+in channels. The physiological role therefore rests mainly on gain-of-function overexpression.
5. **Conceptually corroborated.** The broader principle of S-type anion channels regulating K^+ channels by protein–protein interaction is independently supported by analogous pairs (SLAH3–GORK/SKOR; ALMT12–SLAC1), lending indirect plausibility but not specific replication.

Curation bottom line: GO:0019870 is defensible as a non-core MF annotation with explicit evidence/qualifier caveats (single study, heterologous oocyte assay + overexpression), but it must not be presented as SLAC1's core function, which remains S-type anion (Cl^-/malate/NO_3^-) channel activity driving stomatal closure.

Evidence Matrix

Citation	Evidence type	Stance	Claim tested	Key finding	Context	C li
P 27002025 (Zhang 2016)	Interaction (Y2H, split-ubiquitin, co-IP)	Supports	SLAC1 physically binds inward K^+ channels	SLAC1 interacts with KAT1, KAT2, AKT2 but **not** AKT1	Yeast system + oocyte co-IP	M si d c (I
P 27002025 (Zhang 2016)	Direct assay (Xenopus oocyte voltage-clamp)	Supports	Functional inhibition of K^+in	SLAC1 strongly inhibits KAT1 and AKT2 K^+in; **does not inhibit KAT2**; K^+out unaffected	Heterologous oocytes, overexpression	M h n st h o
P 27002025 (Zhang 2016)	Mutant / gain-of-function + guard-cell patch clamp	Supports (qualified)	In-planta relevance	SLAC1 (WT + truncated) overexpression impairs guard-cell K^+in and light-induced opening; drought up-regulates SLAC1/ SLAH3	Arabidopsis guard cells	L m g fu lo fu o a p
P 23126621 (Laanemets 2013)	Mutant phenotype + guard-cell patch clamp	**Competing / qualifies**	Does *slac1* loss enhance K^+in?	*slac1* **reduces** K^+in and slows opening; rescued by lowering $[Ca^{2+}]$cyt — a Ca^{2+}-sensitivity mechanism, not relief of direct inhibition	Intact Arabidopsis, guard cells	H p d c d in in
P 36653170 (Liu 2023)	Direct assay (oocyte) + mutant	Corroborates concept	Do S-type anion channels regulate K^+	SLAH3 interacts with and partially inhibits **outward** K^+ channels GORK/SKOR	Arabidopsis / oocytes	In la su g m n

Citation	Evidence type	Stance	Claim tested	Key finding	Context	C li
			channels generally?	(different channels)		K s
P 41022577 (Lin 2025)	Direct assay (oocyte + guard cell)	Corroborates concept	Channel–channel regulation of guard-cell anion channels	ALMT12 interacts with and inhibits SLAC1	Arabidopsis guard cells	I s c c b d p d
P 38985761 (Qin 2024)	Structural / electrophysiology	Neutral/ qualifies	SLAC1 activation mechanism	SLAC1 N/C cytosolic regulatory domain autoinhibits the TMD; phosphoactivation releases it	Structure/ AlphaFold + oocyte	I t te i K a S a – a f te
P 24033256 (Acharya 2013)	Interaction + phenotype	Context	OST1 ties anion and K^+in channels	OST1 physically interacts with both SLAC1 and KAT1 and inhibits K^+in	Arabidopsis guard cells	S a p (k m K
IntAct (Q9LD83), this run	Database	Qualifies (absence)	Independent interaction curation	19 curated SLAC1 interactions, all kinases/ phosphatases (CPK6/21/23, PP2CA, SnRK2s/	Database query (this run)	T i n i c

Citation	Evidence type	Stance	Claim tested	Key finding	Context	C li
				OST1); **none with KAT1/AKT2/KAT2**		a d
UniProt/ QuickGO (Q9LD83), this run	Database	Qualifies (current state)	Is GO:0019870 currently annotated?	**No** K^+ channel inhibitor/ regulator MF term present; current MF = anion channel + kinase/ phosphatase binding; BP already has regulation of stomatal opening (GO:1902456)	TAIR/UniProt/ InterPro	R a n

GO Curation Implications

- **Current state (verified this run):** GO:0019870 is **not** currently annotated to SLAC1 in UniProt/TAIR/QuickGO. The decision is therefore whether to **ADD** a new MF term on single-study evidence — a higher bar than retaining an existing one.
- **Term GO:0019870 (potassium channel inhibitor activity, MF):** *Supportable only as a cautious, explicitly-caveated non-core lead.* There is direct functional evidence (oocyte voltage-clamp inhibition of KAT1 and AKT2) plus physical interaction. Appropriate evidence code would be IDA (functional, from heterologous assay) optionally paired with IPI (physical interaction). **Mandatory caveats**: single study, single lab, heterologous oocyte system, gain-of-function overexpression, no independent replication, no database curation of the interaction, and a directionally opposite in-planta loss-of-function phenotype. Note it does **not** inhibit KAT2 despite binding it. A conservative curator could reasonably **decline the addition** pending independent replication.
- **Do not treat as core function.** SLAC1's primary MF is S-type anion channel activity (voltage-gated anion/chloride channel; the gene drives stomatal *closure*). The K^+-channel-inhibitor activity is a secondary/regulatory "moonlighting" activity. If the review is deciding

whether GO:0019870 should be a prominent/core term, the answer is **no — keep it, but mark non-core**.

- **Alternative/parent term consideration:** GO:0015459 (potassium channel regulator activity) is the more conservative parent if a curator judges "inhibitor" too strong given the single-study, heterologous basis. Given the data *do* show inhibition (not activation), GO:0019870 is biologically correct; the conservatism concern is about evidential strength, not direction.
- **Associated BP:** negative regulation of stomatal opening / regulation of stomatal movement is reasonable but is largely a downstream/physiological outcome and is confounded by the Laanemets loss-of-function result.

Mechanistic Scope

- **Immediate molecular activity being tested:** SLAC1 protein binding to Shaker inward-K^+ channel subunits (KAT1, AKT2) and reduction of their K^+in conductance — i.e., a channel–channel inhibitory protein–protein interaction.
- **Directly supported (in the one study):** physical interaction (Y2H/co-IP) and current inhibition in oocytes.
- **Downstream / inferred, not the molecular activity itself:** suppression of light-induced stomatal opening; drought/ABA physiology. These are pathway-level consequences shown mainly by overexpression and are confounded in loss-of-function by Ca^{2+}-sensitivity feedback.
- **Selectivity note:** binding ≠ inhibition. SLAC1 binds KAT2 in Y2H yet does not inhibit it; this dissociation argues for a specific functional, not merely steric/co-localization, effect on KAT1/AKT2 — a point in the hypothesis's favor.

Conflicts and Alternatives

1. **Directional contradiction (Laanemets 2013,** P 23126621 **).**
   Loss of SLAC1 lowers, not raises, K^+in. Explained by elevated cytosolic [Ca^{2+}] increasing

Ca^{2+}-sensitivity-mediated K^+in inhibition — an **alternative mechanism** that does not require direct SLAC1–KAT1 contact.

2. **Kinase-route alternative (Acharya 2013,** **).**
   P 24033256
   OST1 interacts with both SLAC1 and KAT1 and inhibits K^+in; K^+in suppression during closure may be dominated by kinase signaling rather than direct anion-channel binding.
3. **Autoinhibition reinterpretation (Qin 2024,** **).**
   P 38985761
   The SLAC1 N/C termini that Zhang 2016 implicate in KAT1 binding are the same cytosolic regulatory domains that mediate SLAC1 *autoinhibition*; constructs/overexpression effects attributed to KAT1 binding deserve scrutiny.
4. **Heterologous-system artifact.** Oocyte overexpression of a membrane protein can non-specifically depress co-expressed channel currents (membrane crowding, trafficking competition); KAT2 serving as a negative control mitigates but does not eliminate this.
5. **Paralog/term carry-over risk:** SLAH3 (SLAC1 homolog) inhibits different K^+ channels (GORK/SKOR). Curators should avoid propagating GO:0019870 across the SLAC/SLAH family by homology — the specific K^+ targets differ per paralog.

Knowledge Gaps

- **Independent functional replication (highest priority).** Checked: 45 citing papers; none reproduce SLAC1→KAT1/AKT2 patch-clamp inhibition independently. Resolution: an independent guard-cell patch-clamp or oocyte study.
- **Clean loss-of-function test.** Does *slac1* enhance K^+in once the Ca^{2+} feedback is clamped out (Laanemets' <10 nM [Ca^{2+}] condition)? This is the decisive in-planta test of direct inhibition and has not been reported.
- **Native interaction evidence.** IntAct has no SLAC1 records; no reported endogenous co-IP from guard-cell tissue (as opposed to oocyte overexpression).
- **AKT2 specificity.** Full-text confirms AKT2 K^+in inhibition (Suppl. Fig. 2), but this rests on one supplemental figure; independent confirmation absent.
- **Structural basis.** No structure of a SLAC1–KAT1 complex; unknown whether binding occludes the pore/voltage sensor or alters gating/trafficking.

Discriminating Tests

1. **Ca^{2+}-clamped *slac1* patch clamp:** measure guard-cell K^+in in *slac1* vs WT with cytosolic [Ca^{2+}] buffered <10 nM. Direct-inhibition model predicts *higher* K^+in in *slac1*; Ca^{2+}-feedback model predicts rescue to WT levels.
2. **Non-conducting SLAC1 point mutants:** test whether a pore-dead SLAC1 (anion transport abolished) still inhibits KAT1 — separates the inhibitor activity from anion-channel function.
3. **Interface-mutant SLAC1:** map the KAT1-binding surface and test a binding-dead, transport-competent SLAC1 for loss of K^+in inhibition while retaining closure.
4. **Endogenous proximity assay:** split-luciferase/BiFC or co-IP from native guard cells (not overexpression) ± drought to show the interaction occurs at physiological levels.
5. **Quantitative stoichiometry:** compare SLAC1:KAT1 abundance in guard cells to judge whether direct binding can plausibly titrate KAT1 in planta.

Curation Leads (require curator verification)

- **Lead 1 — Retain GO:0019870 as a non-core MF annotation** with an explicit caveat that evidence is from a single study using heterologous oocyte voltage clamp plus guard-cell overexpression (IDA), supported by IPI (Y2H/co-IP), and is not independently replicated. Verify snippet: *"Voltage-clamp results demonstrated that SLAC1 inhibited KAT1 dramatically, but did not inhibit KAT2."* (P 27002025).
- **Lead 2 — Add a NOT/contrast note on specificity:** SLAC1 binds KAT2 (Y2H) but does **not** inhibit it; do not annotate KAT2 inhibition. Verify snippet: full-text "SLAC1 Strongly Inhibits KAT1, but Not KAT2, in Xenopus laevis Oocytes."
- **Lead 3 — Flag competing mechanism:** cite P 23126621 as a qualifier that in-planta *slac1* K^+in reduction is Ca^{2+}-mediated, not a relief-of-direct-

inhibition effect. Verify snippet: *"severely reduced activity of inward K+ (K+in) channels in slac1 guard cells."*

- **Lead 4 — Do not homology-propagate** GO:0019870 to SLAH3/other SLAC paralogs; SLAH3's K^+ targets (GORK/SKOR, outward) differ (P 36653170).
- **Candidate alternative term if curator deems "inhibitor" too strong:** GO:0015459 potassium channel regulator activity (direction still supports "inhibitor").
- **Suggested curator question:** Is a single-lab, heterologous + overexpression result sufficient for an experimental MF annotation given the opposite-direction loss-of-function phenotype? If policy requires independent corroboration for core terms, keep GO:0019870 as non-core only.

Supported vs Refuted Clauses of the Seed Hypothesis

Clause	Status
SLAC1 physically interacts with KAT1 and AKT2 (not AKT1)	**Supported** (Zhang 2016; single study; not in IntAct)
SLAC1 directly inhibits KAT1 and AKT2 K^+in	**Supported in vitro** (oocyte; single study)
SLAC1 does not inhibit KAT2	**Supported** (binds but does not inhibit)
Inhibition contributes to suppression of stomatal opening in planta	**Partially supported** (gain-of-function only; loss-of-function is opposite, Ca^{2+}-confounded)
Independently replicated or corroborated	**Not met** (concept corroborated via analogues; SLAC1–KAT1 not independently replicated)
Not contradicted by later work	**Partially contradicted / qualified** (Laanemets 2013 directional conflict; Qin 2024 autoinhibition reinterpretation)

Analyses run across iterations: NCBI E-utilities (efetch/elink/esummary) for the reference paper, full-text AKT2/patch-clamp verification from PMC4863386, cited-by screen of all 45 papers citing Zhang 2016, IntAct web-service interactor query for Q9LD83 (19 interactors, all kinases/phosphatases), and UniProt/QuickGO MF-annotation checks confirming GO:0019870 is not currently annotated.

Provenance artifacts (computed this run): `evidence_matrix.csv`, `go_decision_table.csv`, `hypothesis_clause_verdicts.csv` in the job directory.
